A. V. PAREKH TECHNICAL INSTITUTE, RAJKOT
COMPUTER ENGINEERING DEPARTMENT

Unit – IV

Cloud Computing
(Course Code: 4360709)

Virtualization

Vision Mission

Vision of Computer Department

Develop globally competent Computer Engineering Professionals who achieve excellence in an environment conducive for technical knowledge, skills, moral and ethical values with a focus to serve the society.

Mission of Computer Department

M1 : To provide state of the art infrastructure and facilities for imparting quality education and computer engineering skills for societal benefit.

M2 : Adopt industry oriented curriculum with an exposure to technologies for building systems & application in computer engineering.

M3 : To provide quality technical professional as per the industry and societal needs, encourage entrepreneurship, nurture innovation and life skills in consonance with latest interdisciplinary trends.

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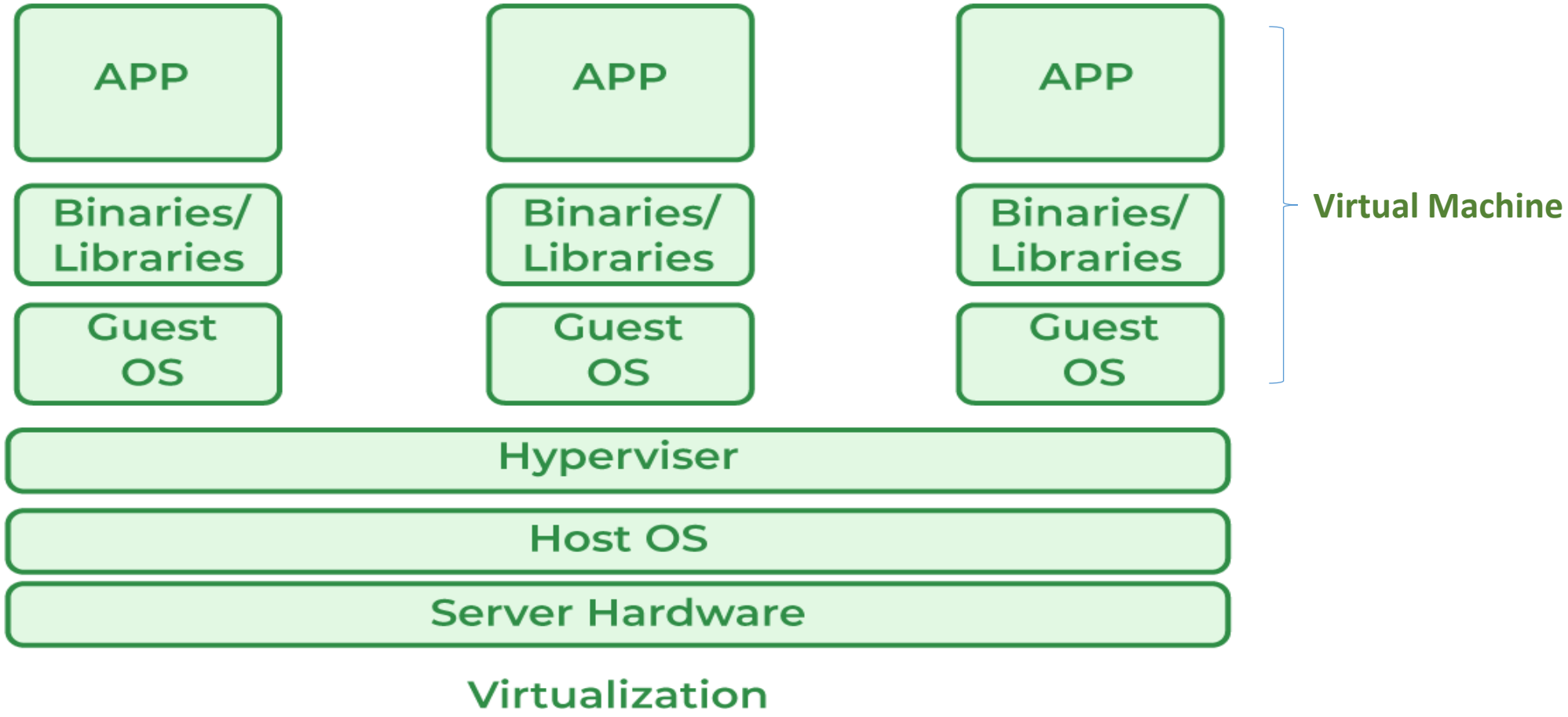
Introduction of Virtualization

- **Virtualization** is the "creation of a virtual (rather than actual) version of something, such as a server, a desktop, a storage device, an operating system or network resources".
- Formally, Virtualization is a technique, which allows sharing of a single physical instance of a resource or an application among multiple customers and organizations.
- This is done by assigning a logical name to a physical storage and providing a pointer to that physical resource when demanded.

- A Virtual machine provides an environment that is logically separated from the underlying hardware.
- The machine on which the virtual machine is going to create is known as **Host Machine** and that virtual machine is referred as a **Guest Machine.**



Levels of Implementing Virtualization

1. Instruction Set Architecture Level (ISA)
2. Hardware Abstraction Level (HAL)
3. Operating System Level
4. Library Level
5. Application Level

Five Levels of Virtualization

Application Level

JVM / .NET CLR

Library Level

WINE / vCUDA

Operating System Level

Virtual Environment / FVM

Hardware Abstraction Level

VMWare / Virtual PC

Instruction Set Architecture Level

BIRD / Dynamo

Instruction Set Architecture Level (ISA)

In ISA, virtualization works through an ISA emulation. This is helpful to run heaps of legacy code which was originally written for different hardware configurations.

- These codes can be run on the virtual machine through an ISA.
- A binary code that might need additional layers to run can now run on an x86 machine or with some tweaking, even on x64 machines. ISA helps make this a hardware-agnostic virtual machine.
- The basic emulation, though, requires an interpreter. This interpreter interprets the source code and converts it to a hardware readable format for processing.

Hardware Abstraction Level (HAL)

- As the name suggests, this level helps perform virtualization at the hardware level. It uses a bare hypervisor for its functioning.
- This level helps form the virtual machine and manages the hardware through virtualization.
- It enables virtualization of each hardware component such as I/O devices, processors, memory, etc.
- This way multiple users can use the same hardware with numerous instances of virtualization at the same time.

Operating System Level

- At the operating system level, the virtualization model creates an abstract layer between the applications and the OS.
- It is like an isolated container on the physical server and operating system that utilizes hardware and software. Each of these containers functions like servers.
- When the number of users is high, and no one is willing to share hardware, this level of virtualization comes in handy.
- Here, every user gets their own virtual environment with dedicated virtual hardware resources. This way, no conflicts arise.

Library Level

- OS system calls are lengthy and cumbersome. Which is why applications opt for APIs from user-level libraries.
- Most of the APIs provided by systems are rather well documented. Hence, library level virtualization is preferred in such scenarios.
- Library interfacing virtualization is made possible by API hooks. These API hooks control the communication link from the system to the applications.
- Some tools available today, such as vCUDA and WINE, have successfully demonstrated this technique.

Application Level

- Application-level virtualization comes handy when you wish to virtualize only an application. It does not virtualize an entire platform or environment.
- On an operating system, applications work as one process. Hence it is also known as process-level virtualization.
- It is generally useful when running virtual machines with high-level languages. Here, the application sits on top of the virtualization layer, which is above the application program.
- The application program is, in turn, residing in the operating system.
- Programs written in high-level languages and compiled for an application-level virtual machine can run fluently here.

Types of Virtualization

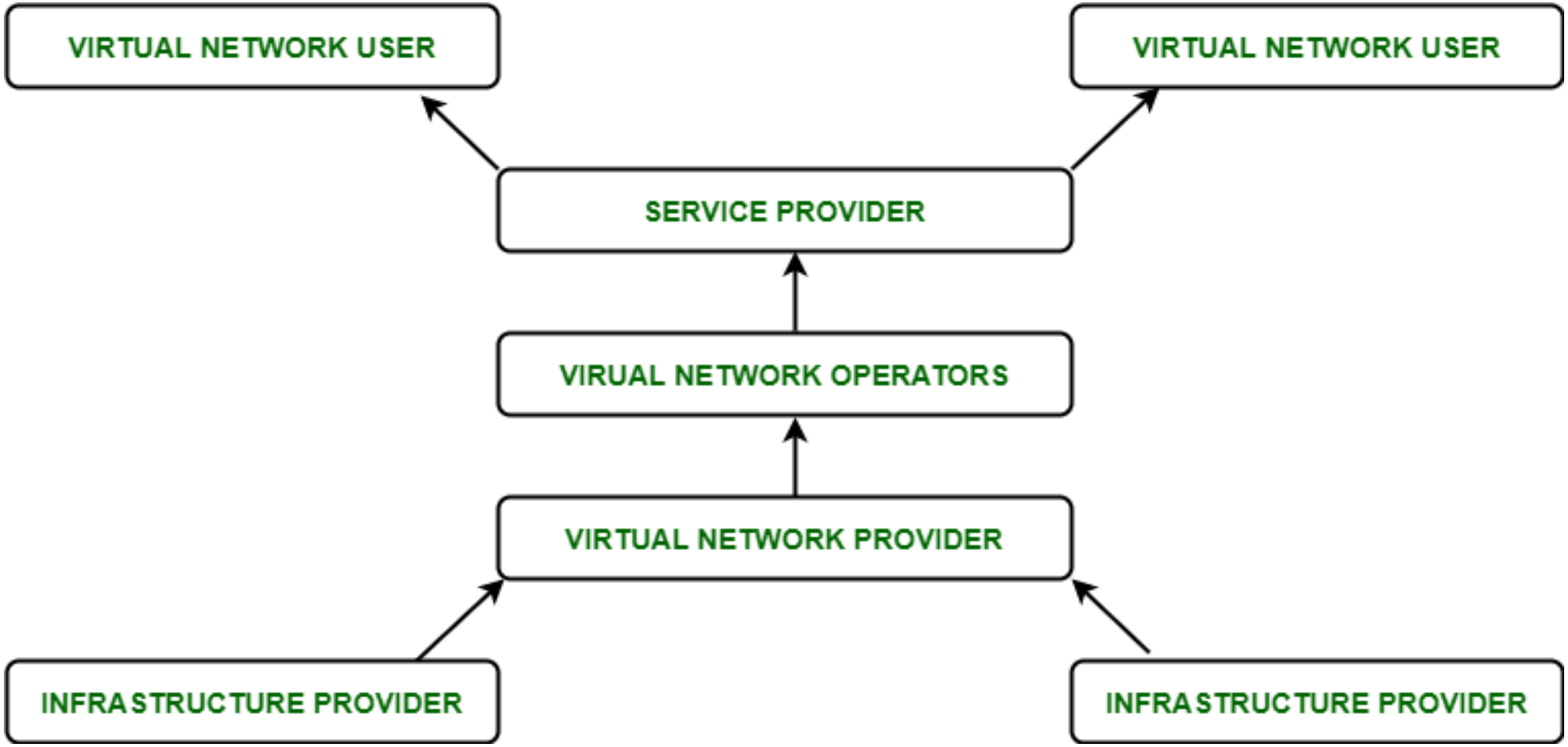
1. Desktop virtualization
2. Network Virtualization
3. Storage Virtualization
4. Data virtualization
5. Hardware Virtualization
6. Software Virtualization

Desktop virtualization

- Desktop virtualization allows the users’ OS to be remotely stored on a server in the data center.
- It allows the user to access their desktop virtually, from any location by a different machine.
- Users who want specific operating systems other than Windows Server will need to have a virtual desktop.
- The main benefits of desktop virtualization are user mobility, portability, and easy management of software installation, updates, and patches.

Network Virtualization,

- The ability to run multiple virtual networks with each having a separate control and data plan.
- It co-exists together on top of one physical network. It can be managed by individual parties that are potentially confidential to each other.
- Network virtualization provides a facility to create and provision virtual networks, logical switches, routers, firewalls, load balancers, Virtual Private Networks (VPN), and workload security within days or even weeks.


VIRTUAL NETWORK USER
VIRTUAL NETWORK USER
SERVICE PROVIDER
VIRUAL NETWORK OPERATORS
VIRTUAL NETWORK PROVIDER
INFRASTRUCTURE PROVIDER
INFRASTRUCTURE PROVIDER

Storage Virtualization

- Storage virtualization is an array of servers that are managed by a virtual storage system.
- The servers aren’t aware of exactly where their data is stored and instead function more like worker bees in a hive.
- It makes managing storage from multiple sources be managed and utilized as a single repository.
- Storage virtualization software maintains smooth operations, consistent performance, and a continuous suite of advanced functions despite changes, breaks down, and differences in the underlying equipment.

Data virtualization

- This is the kind of virtualization in which the data is collected from various sources and managed at a single place.
- It is not known more about the technical information like how data is collected, stored & formatted then arranged that data logically so that its virtual view can be accessed by its interested people and stakeholders, and users through the various cloud services remotely.
- Many big giant companies are providing their services like Oracle, IBM, At scale, Cdata, etc.

Hardware Virtualization

- When the virtual machine software or virtual machine manager *(VMM) is directly installed on the hardware system* is known as hardware virtualization.
- The main job of hypervisor is to control and monitoring the processor, memory and other hardware resources.
- After virtualization of hardware system we can install different operating system on it and run different applications on those OS.
- Its mostly implemented for the server platforms, because controlling virtual machines is much easier than controlling a physical server.

Software Virtualization

- *Software Virtualization is able to abstract the software installation procedure and create virtual software installations*.
- **Virtualized software** is an application that will be "installed" into its own self-contained unit.
- Example of software virtualization is *VMware software, virtual box* etc.

Benefits of Virtualization

- More flexible and efficient allocation of resources.
- Enhance development productivity.
- It lowers the cost of IT infrastructure.
- Remote access and rapid scalability.
- High availability and disaster recovery.
- Pay peruse of the IT infrastructure on demand.
- Enables running multiple operating systems.

Drawback of Virtualization

- **High Initial Investment:** Clouds have a very high initial investment, but it is also true that it will help in reducing the cost of companies.
- **Learning New Infrastructure:** As the companies shifted from Servers to Cloud, it requires highly skilled staff who have skills to work with the cloud easily, and for this, you have to hire new staff or provide training to current staff.
- **Risk of Data:** Hosting data on third-party resources can lead to putting the data at risk, it has the chance of getting attacked by any hacker or cracker very easily.

Characteristics of Virtualization

- **Increased Security:** The ability to control the execution of a guest program in a completely transparent manner opens new possibilities for delivering a secure, controlled execution environment. All the operations of the guest programs are generally performed against the virtual machine, which then translates and applies them to the host programs.
- **Managed Execution:** In particular, sharing, aggregation, emulation, and isolation are the most relevant features.
- **Sharing:** Virtualization allows the creation of a separate computing environment within the same host.
- **Aggregation:** It is possible to share physical resources among several guests, but virtualization also allows aggregation, which is the opposite process.

Applications of Virtualization

- Data-integration
- Business-integration
- Service-oriented architecture data-services
- Searching organizational data

Technologies of virtualization : Hypervisors

- A hypervisor, also known as a virtual machine monitor or VMM.
- The hypervisor is a piece of software that allows us to build and run virtual machines.
- The guest VMs are independent of the host hardware which is one of the major benefits of the Hypervisor.
- Thus they can be quickly switched between servers.
- It allows several virtual machines to operate on a single physical server.

Types of hypervisors

- There are two types of hypervisors: "Type 1" (also known as "bare metal") and "Type 2" (also known as "hosted").
- A type 1 hypervisor functions as a light operating system that operates directly on the host's hardware, while a type 2 hypervisor functions as a software layer on top of an operating system, similar to other computer programs.

Type 1 hypervisor

- Type 1 hypervisor replaces the host operating system, and the hypervisor schedules VM services directly to the hardware.
- The type 1 hypervisor is very much commonly used in the enterprise data center or other server-based environments.
- It includes KVM, Microsoft Hyper-V, and VMware vSphere. If we are running the updated version of the hypervisor then we must have already got the KVM integrated into the Linux kernel in 2007.

Type 2 hypervisor

- It is also known as a hosted hypervisor, The type 2 hypervisor is a software layer or framework that runs on a traditional operating system.
- It operates by separating the guest and host operating systems. The host operating system schedules VM services, which are then executed on the hardware.
- Individual users who wish to operate multiple operating systems on a personal computer should use a form 2 hypervisor.
- This type of hypervisor also includes the virtual machines with it.

Comparison on Type-1 and Type-2 Hypervisor

- Hardware acceleration technology improves the processing speed of both bare-metal and hosted hypervisors, allowing them to build and handle virtual resources more quickly.
- A type 1 hypervisor functions as a light operating system that operates directly on the host's hardware, while a type 2 hypervisor functions as a software layer on top of an operating system, similar to other computer programs.
- Bare-metal hypervisors are isolated from the attack-prone operating system thus they are extremely stable.
- Bare-metal hypervisors are usually faster and more powerful than hosted hypervisors. For these purposes, the majority of enterprise businesses opt for bare-metal hypervisors for their data center computing requirements.

- Hosted hypervisors run inside the OS, they can be topped with additional (and different) operating systems.
- The hosted hypervisors have longer latency than bare-metal hypervisors which is a very major disadvantage of the it. This is due to the contact between the hardware and the hypervisor must go through the OS's extra layer.

Choosing the right hypervisor (Xen, KVM)

- Xen is an open source hypervisor that runs directly on the system hardware.
- It's a type 1 hypervisor, also known as a bare-metal hypervisor that creates logical pools of system resources so that many virtual machines can share the same physical resources.
- Xen inserts a virtualization layer between the system hardware and the virtual machines.
- This allows Xen to dynamically allocate the system hardware to any guest operating system.

KVM (Kernel-based **Virtual** Machine)

- Kernel-based Virtual Machine (KVM) is an open source virtualization technology built into Linux®. Specifically, KVM lets you turn Linux into a hypervisor that allows a host machine to run multiple, isolated virtual environments called guests or virtual machines (VMs)
- KVM provides high performance and scalability due to its native support for hardware virtualization extensions and paravirtualization drivers. Furthermore, KVM has an abundant ecosystem of management tools, including libvirt, virt-manager, virsh, and more.
- KVM converts Linux into a type-1 (bare-metal) hypervisor. All hypervisors need some operating system-level components—such as a memory manager, process scheduler, input/output (I/O) stack, device drivers, security manager, a network stack, and more—to run VMs. KVM has all these components because it’s part of the Linux kernel. Every VM is implemented as a regular Linux process, scheduled by the standard Linux scheduler, with dedicated virtual hardware like a network card, graphics adapter, CPU(s), memory, and disks.

How do you choose between KVM and Xen?

- When selecting a hypervisor for Linux virtualization, there is no definitive answer as it depends on your use case, requirements, and preferences.
- However, KVM is a good choice if you want to leverage the Linux kernel features for virtualization and containers, run a wide range of guest operating systems, use various management tools and plugins, and achieve high performance and scalability.
- Alternatively, Xen is a good option if you’re looking for a bare-metal hypervisor with high isolation and security, reduced overhead and complexity, and a mature and stable code base.

Queries ???